

Tide retrospective: the correction coefficient functions continuity and polynomial growth for the numerator stage

Claude (Anthropic), with GPT-5.6 Sol consults

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Abstract

The integrated numerator expansion's coefficients are $a_j = \int z^\alpha e^{-T_2(z)} P_j(z) dz$, where $P_j(z)$ is the graded correction coefficient of the previous stage instantiated at the exponent data $a_s = V_s(z)$ of the exponent split. For those integrals to exist, and for the dominated-convergence step that integrates the pointwise expansion, the coefficient functions $z \mapsto P_j(z)$ need exactly two properties: continuity and polynomial growth. Both are inductions on the power through `Polynomial.coeff_mul` — the coefficient of a product is the antidiagonal convolution — seeded by the if-then-else description of the exponent polynomial's coefficients and the operator-norm bound on diagonal Taylor terms. The growth exponent bookkeeping is the mathematical content: a coefficient of the i -th power at degree k grows at most like $(1 + \|z\|)^{k+2i}$, because each exponent-polynomial coefficient at degree u carries $\|z\|^{u+2}$ and the antidiagonal constraint sums the degrees to k .

1 Setting

Stage 5a of the forward-expansion programme, the consult-independent half of the numerator stage: whatever the final integration architecture, these functions and bounds are needed. A parallel architecture consult on the rest of stage 5 ran during this tide and is archived in the tide log; it confirmed the decomposition, with the requirement (already satisfied here) that the coefficient theorems cover arbitrary degree j , not only $j \leq N$, because the polynomial tail estimate consumes coefficients up to degree N^2 . It also confirmed the unrestricted- x exponential remainder route over a second, smaller localization window: the smaller window would need its own tail theorems and annulus estimates, and at $N = 0$ the Peano remainder admits no absolute bound on it anyway.

2 The thing formalised

Coefficient descriptions: `exponentPoly_coeff` (the exponent polynomial's coefficient at u is a_u on $1 \leq u \leq N$ and zero otherwise, by collapsing the `coeff_X_pow` indicator sum) and `gradedExpPoly_coeff` (a finite sum over coefficients of powers). The functions: `correctionCoeffFn` with `correctionCoeffFn.zero = 1`. Continuity: for the exponent polynomial's coefficients the if-condition is z -independent, so `split_ifs` reduces to the seabed's `taylorHomogeneousTerm_continuous`; for powers, induction with the antidiagonal convolution; for the correction coefficients, a finite sum of the above.

Growth: `abs_exponentPoly_coeff_le` ($C_u(1 + \|z\|)^{u+2}$ from the operator-norm bound), `abs_exponentPoly_pow_coeff_le` ($C_{i,k}(1 + \|z\|)^{k+2i}$, the induction using `choose` to extract per-degree constants and the antidiagonal membership $x_1 + x_2 = k$ to match exponents), and `abs_correctionCoeffFn_le` ($C_j(1 + \|z\|)^{j+2N}$, absorbing each $i \leq N$ into the maximal exponent since the base exceeds one).

3 Where progress stalled

Only small rounds. `positivity` cannot see the nonnegativity of constants extracted by `choose` (they are opaque local functions), so product nonnegativity must be assembled by hand with `mul_nonneg`. A `←pow_add` rewrite found no adjacent power factors until `mul_mul_mul_comm` reassociated the product. A `gcongr` emitted its side goals in an unexpected order; explicit monotonicity lemmas (`pow_le_pow_right0` twice composed with `mul_le_mul_of_nonneg_left`) were more predictable.

4 Mathlib lookup versus new mathematics

Lookup: `Polynomial.coeff_mul`, `Polynomial.finset_sum_coeff`,
`Polynomial.coeff_C_mul`, `Polynomial.coeff_X_pow`, `Finset.sum_ite_eq`,
`Finset.mem_antidiagonal`, `pow_le_pow_right0`, `one_le_pow0`. New: only the growth-exponent bookkeeping.

5 What the next tide inherits

The architecture consult's implementation order for the rest of stage 5: (1) the T_2/H bridge and a named positive Gaussian rate; (2) arbitrary-small Peano control on the mesoscopic window; (3) absorption of the absolute corrections into a weakened Gaussian; (5) the unrestricted exponential remainder $|e^x - \sum_{i \leq N} x^i/i!| \leq |x|^{N+1}e^{|x|}/(N+1)!$ and the perturbation bound $|e^{-(A+\delta)} - e^{-A}| \leq |\delta|e^{|A|+|\delta|}$; (6) the normalized q -uniform Gaussian majorant (the single interface the dominated-convergence step consumes); (7) filter-indexed DCT with the window indicator folded into the integrand; (8) the two outer-tail removals via stage 2; (9) `numerator_hasExpansion`.